

human intervention is minimised. For example, M2M systems controlling the ventilation and fire systems are set in such a manner that they work optimally at all times.

Architects have started using sensors that capture site-specific data and send it back to the office where the analytic software helps to maximise the potential efficiency of a building in blueprint drafting. M2M can reduce CO₂ emissions by as much as 1.6Gt by 2020 enhancing the energy efficiency of building systems which include heating, cooling, ventilation, lighting, electronics and appliances and security systems.

Agriculture. Lately, we have been seeing deviations in climate and unexpected floods and typhoons. Our agricultural sector is threatened by these and thus demands a change in the manner we grow, harvest, transport and store food.

M2M can help bring down the methane emissions which are produced from cattle digestion (enteric fermentation) by monitoring cattle health and optimising their grazing patterns.

Remote monitoring of soil conditions, smart farming and smart watering are some other ways by which M2M technologies can help reduce GHG emissions. These come under the genre of 'precision agriculture.' For example, tractors that have M2M-enabled auto-steer technologies can be mapped with high precision in terms of their location in the field. They can then automatically reference that data with information on crop aspects, such as soil and water

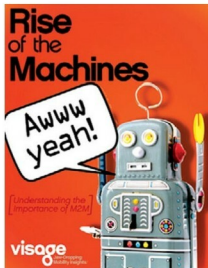
quality from other sensors and use water and fertilisers very accurately over each square metre of the field with almost zero wastage.

There are also savings in fuel consumption as tractors are used only where and when they are really required. A great example of M2M technology at work in the farms in India is NanoGanesh—a device that enables farmers to switch on irrigation pumps used for watering crops in remote locations using their mobile phones along with a mobile modem that attaches to the starter of the irrigation pump. This is a big boon for the farmers who had to walk several kilometres to their fields to switch on or switch off their water pumps. Nano Ganesh allows farmers to remotely check to see that there is electricity, and to automatically turn the pump on or off.

It is estimated by BCG & GeSI that M2M could help reduce another 1.6Gt of CO₂ emissions by 2020 in the agriculture sector by reducing deforestation, managing livestock and improving the efficiency of planting, seeding, harvesting, fertiliser application and water use. This will enable a larger amount of crops to be grown using fewer resources and in saving money for our farmers.

Endpoint

With time, more and more things in our daily lives will become connected—our home appliances, houses, cars and offices. And the more they will be able to communicate with each other. We can look forward to a whole new and completely connected world.



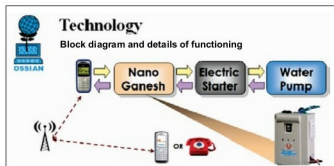
Some day machines will verbally communicate with humans and other machines (Source: <http://get.visagemobile.com>)

be 12.5 billion M2M devices globally by 2020. Another transformation that we will see is the shift in the distribution of M2M applications.

In the early days, the M2M apps were mostly centred on the automotive and transport sectors and were commercial in nature. As the technology developed, applications in other sectors, such as health, agriculture and energy, also became prominent. However, their usage was more concentrated in the developed nations.

In time to come, M2M apps will dominate all sectors and include several other sectors, such as banking and real estate. We will have smart homes, smart clothing, smart factories, smart cities and smart anything that you can think of. These apps will be both commercial and consumer in nature and touch several people in emerging and developing countries too.

The popular science-fiction film Star Wars exposed a world where machines verbally communicated with humans and other machines. In the years gone by, it seemed the technological wonders depicted were too far-fetched to ever resemble reality. However, some of these once-futuristic concepts would soon become reality. ●



How Nano Ganesh works (Source: <http://www.technospot.in>)

As per the 2013 report Machine to Machine Technologies: Unlocking the potential of a \$1 trillion Industry By 2020, produced by the Carbon War Room Research and Intelligence Group, there will

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